

SATYAM KUMAR

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EXPERIENCE

Tata 1mg

Gurugram, India

Senior Software Development Engineer (SDE III) – AI Systems & Platform

April 2026 – Present

- Authored a persistent multi-agent orchestration framework (extending PydanticAI) providing durable agent/session persistence, resumable and branchable conversation threads, token-aware context compaction, and full LLM observability – adopted as the core orchestration layer across **6+** internal AI products
- Built long-lived autonomous coding-agent infrastructure supporting asynchronous execution, real-time progress streaming, pause/resume checkpoints, and distributed cancellation for complex multi-step coding tasks
- Extended **DeputyDev**'s code-intelligence layer: Tree-sitter-based language-aware parsing/chunking, semantic vector search with symbol/function/class-level retrieval and reranking, and multi-provider LLM routing (OpenAI, Anthropic, Google, others) with automatic fallback across a compiled local binary, backend services, and VS Code extension
- Led cross-functional adoption of GenAI tooling across 1mg's engineering org, improving developer throughput by **30%+**; co-authored a published study measuring the platform's real-world productivity impact (see Publications)
- Designed an AI-assisted catalog-management system spanning 40+ entity types with mandatory human-in-the-loop approval and full version/audit history, ensuring AI-generated changes never write directly to production data
- Built **1mg MCP**, an authenticated MCP server exposing commerce tools (product search, cart, checkout, order tracking) with location-aware context enforcement, enabling MCP-compatible AI clients to transact on the platform
- Built an MCP-based agent-memory platform with scoped user/repo/org memory and human-review gates for shared writes; also shipped a LangGraph-based healthcare AI assistant for lab-test discovery and booking

Tata 1mg

Gurugram, India

Software Development Engineer II

April 2024 – April 2026

- Built and led **DeputyDev**, an AI-powered developer platform (code generation, code review, productivity analytics) from inception to **300+ engineers** deployed org-wide, distributed via the VS Code Marketplace with multimodal understanding and codebase-aware agentic workflows
- Overhauled query-solving, history, and intent-resolution systems, improving response speed **3–4x**, reliably handling **1,000+ daily queries**, and cutting chat error rates from **300+ to under 100/day**
- Re-architected the payments/refunds pipeline (**50%** less memory, **30%** lower latency) and refactored the Python backend (Astral + UV + Ruff), cutting setup time from hours to minutes
- Enhanced Care Plan and Loyalty systems, driving **6x** adoption growth (200 → 1,200 orders/day) and cutting third-party integration time by **30x**

Tata 1mg

Gurugram, India

Software Development Engineer I

October 2022 – April 2024

- Enhanced the loyalty and rewards service for fault tolerance across high-traffic order flows
- Created a system to onboard third-party partnerships via simple APIs, saving **30x** implementation time

Yellow Class (now SpeakX)

Remote (later Gurugram), India

Software Development Engineer

July 2021 – October 2022

- Built microservices to improve performance and scale for a top-10 Indian education app
- Developed the communications backend, enabling **5x** growth in user interactions
- Created the Quiz module from scratch, handling **1,000+ requests/min** with minimal resource use
- Designed a URL shortener with caching and message queues for low-latency performance
- Maintained libraries, introduced strict typing, and built a reusable nested-discriminator library in Mongoose
- Implemented backend for mini-games, collectibles, and rewards, boosting engagement and retention

PUBLICATIONS

Intuition to Evidence: Measuring AI's True Impact on Developer Productivity – Co-author. Year-long, 300-engineer enterprise study on AI-assisted development; found a **31.8%** reduction in PR review cycle time and a **28%** increase in production code shipment volume. *arXiv:2509.19708, 2025.*

OPEN SOURCE

Google Summer of Code – Kiwix | *Python, Offliners, Zimfarm*

Jun 2020 – Aug 2020

- Selected among 1,199 global participants; improved Python offliners and Zimfarm to archive web content into ZIM files, achieving **10x** faster offlining and up to **50%** smaller output files across seven projects

OpenEdx ZIM Scraper | *Python (Freelance)*

Aug 2020 – Dec 2020

- Designed an improved OpenEdx scraper to generate ZIM archives from any OpenEdx-powered platform (e.g. edX); contributed to core ZIM-creation libraries

EDUCATION

Indian Institute of Information Technology, Kalyani

B.Tech in Computer Science and Engineering; CGPA: 9.26/10

Kalyani, India

Expected June 2021

AWARDS

2025 – Organizational Department Award for Performance, Tata 1mg (contributions to DeputyDev and GenAI initiatives)

2020 – National Winner, Smart India Hackathon (Problem Statement DR135, Govt. of Goa)

2019 – 2nd Runner-up, CodeUtsava 3.0 Hackathon, NIT Raipur

2018 – Most Innovative Hardware Hack, HACKABIT

2017 – 2nd Rank (National), SOF National Cyber Olympiad

TECHNICAL SKILLS

AI / ML: Multi-agent orchestration, agentic workflows, LLM routing & fallback, RAG, semantic & vector search, prompt engineering & caching, multimodal inference, model evaluation

Frameworks: PydanticAI, LangGraph, Model Context Protocol (MCP), LLM observability (Langfuse)

Languages: Python, JavaScript, TypeScript, SQL

Infrastructure & Data: PostgreSQL, MongoDB, Redis, Kafka, RabbitMQ/SQS, Qdrant, Weaviate, API governance